

Noether Sheet 4

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These were my solutions to the Noether Mentoring Scheme Sheet 4. Problem statements are abridged.

Problem 1

Solve the simultaneous equations

$$\begin{aligned}4x + 3y &= 496 \\ \sqrt{5x} + \sqrt{5y} &= \sqrt{17x + 7y}.\end{aligned}$$

Solution

Define $\alpha = 496$. Square the second equation twice to end up with

$$144x^2 - 52xy + 4y^2 = 0.$$

Now substitute in $4x = \alpha - 3y$ to get

$$9(\alpha - 3y)^2 - 13y(\alpha - 3y) + 4y^2 = 0$$

which simplifies to

$$124y^2 - 67\alpha y + 9\alpha^2 = 0$$

which yields the solutions $y = 144, 124$, with corresponding values of $x = 16, 31$, and both solutions can be checked to work. ■

Problem 2

(a) Show algebraically that

$$\binom{n+1}{r+1} = \binom{n}{r} + \binom{n}{r+1}.$$

(b) i. How many ways are there to choose $r+1$ objects from a set of $n+1$.

ii. Suppose that one of the $n+1$ objects is special. How many ways can you choose the $r+1$ items including the special one?

iii. Hence give a combinatorial proof that

$$\binom{n+1}{r+1} = \binom{n}{r} + \binom{n}{r+1}.$$

(c) Show that

$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n.$$

Solution

(a) The RHS reads

$$\begin{aligned} \binom{n}{r} + \binom{n}{r+1} &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r+1)!(n-r-1)!} \\ &= \frac{n!}{r!} \left(\frac{1}{(n-r)!} + \frac{n-r}{(r+1)(n-r)!} \right) \\ &= \frac{n!}{r!} \left(\frac{(r+1) + (n-r)}{(r+1)(n-r)!} \right) \\ &= \frac{n!}{r!} \left(\frac{n+1}{(r+1)(n-r)!} \right) \\ &= \frac{(n+1)!}{(r+1)!(n-r)!} \\ &= \binom{n+1}{r+1}. \quad \square \end{aligned}$$

(b) i. There are $\binom{n+1}{r+1}$ ways to choose $r+1$ objects from a set of $n+1$.

ii. We must pick the one special object, other than it, we need to pick r items from a set of n items. There are $\binom{n}{r}$ ways to do this.

iii. From i and ii we know that there are $\binom{n+1}{r+1} - \binom{n}{r}$ ways to choose a set of $r+1$ objects from a set of $n+1$, excluding the special one. Alternatively, to pick $r+1$ objects excluding the special one, we can just remove the special object from the set and pick $r+1$ objects from this new set. There are $\binom{n}{r+1}$ ways to do this. But this means that

$$\binom{n+1}{r+1} - \binom{n}{r} = \binom{n}{r+1}$$

which re-arranges to the desired result. $\square$

(c) Note that the sum is exactly $(1 + 1)^n$ due to the binomial theorem which is the same as 2^n . $\blacksquare$

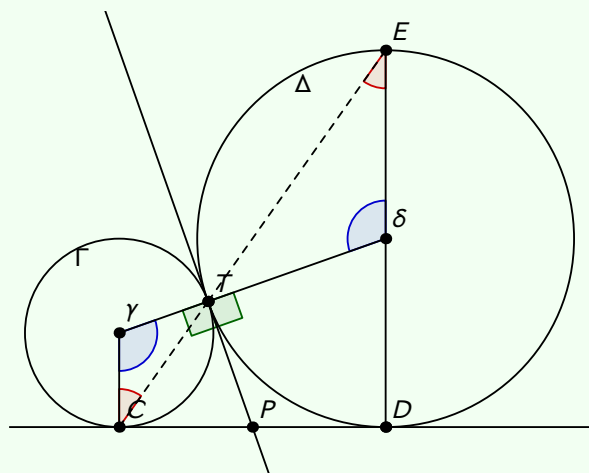
Problem 3

Suppose that l is a line, and that Γ and Δ are circles, tangent to l at C and D respectively and also tangent to each other at T . Let E be the opposite end of the diameter from Δ from D .

Prove that C, T and E are collinear.

Solution

The following diagram gives an overview of our argument.



Claim 1: γ, T and δ are collinear.

Proof: We note that because radii meet tangents at right angles $\angle PT\gamma = \angle PT\delta = 90^\circ$. Now since $\angle \gamma T \delta = 90 + 90 = 180^\circ$, we can conclude that γ, T and δ lie on a common line. $\square$

Claim 2: $\angle C\gamma T = \angle T\delta E$.

Proof: Notice that since both $C\gamma$ and $D\delta$ are perpendicular to the line l , we have $C\gamma \parallel D\delta$. Now due to Alternate angle theorem $\angle C\gamma T = \angle T\delta E$. $\square$

Finally since $\triangle TC\gamma$ and $\triangle TE\delta$ are both isocles, $\angle TC\gamma = \angle TE\delta$. Now since both CT and TE make the the same angle with the with the line perpendicular to l , we have $CT \parallel TE$ and since the two segments both pass through the point T , it is the case that C, T and E are collinear. $\blacksquare$

Problem 4

- (a) What is the remainder when 1000^{1000} is divided by 89?
 (b) What is the remainder when 1000^{1000} is divided by 87?

Solution

(a) We notice that 89 is a prime number. Since $1000 = 11 \cdot 89 + 21$, we wish to compute $21^{1000} \pmod{89}$. We use Fermat's little theorem,

$$21^{1000} \equiv (21^{11})^{89} \cdot 21^{21} \equiv 21^{11+21} \equiv 21^{32} \pmod{89}.$$

Now since $21^2 = 441 \equiv 32 \pmod{89}$ we have to compute $32^{16} \pmod{89}$. Now we can keep applying this same trick to find

$$32^{16} \equiv 45^8 \equiv 67^4 \equiv 39^2 \equiv 8 \pmod{89}. \quad \square$$

(b) We make use of Euler's theorem. Note that $\varphi(87) = 87 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{29}\right) = 56$. Since $3000 = 56 \cdot 53 + 32$ we can write

$$1000^{1000} = 10^{3000} = (10^{53})^{56} \cdot 10^{32}.$$

Because 10 and 87 are coprime we can apply Euler's theorem to reduce the last expression mod 87. It becomes

$$(1)10^{32} \pmod{87}.$$

We can use the same trick as we used in part (a),

$$10^{32} \equiv 13^{16} \equiv 82^8 \equiv 25^4 \equiv 16^2 \equiv 82 \pmod{87}. \quad \blacksquare$$

Problem 5

Jose buys a lottery ticket for his 45-th birthday, for which he has to select six numbers between 1 and 59. To celebrate his age, he chooses the number 45 as one of them. In his six numbers, he uses all the digits 0 to 9 once each, and the product of his numbers is a perfect square.

What numbers did he pick?

Solution

We begin with a few claims.

Claim 1: The chosen number that uses the digit 7 is either 7 or 27.

Proof: Let x be the chosen number that uses the digit 7. If x has two digits, then its tens digit must be 1, 2, 3 since Jose selects numbers from 1 to 59. Note that 17 and 37 are prime numbers.

- If the number 17 is chosen then, there must be another multiple of 17 among the chosen numbers to make the product square. The only other multiples of 17 in the given range are 34, 51. But 34 can't be used since the digit 4 has been taken, and 51 can't be used since the digit 5 has been taken.
- If the number 37 is chosen, then there must be another multiple of 37 among the chosen numbers to make the product square. However since 37 itself is the only multiple of 37 in the given range, the number 37 can't be chosen.

Thus it must be the case that $x \in \{7, 27\}$. $\square$

Claim 2: The chosen number that uses the digit 0 is either 10, 20 or 30.

Proof: Since $45 = 3^2 \cdot 5$, one of the chosen numbers must have an odd 5-adic valuation, in order to ensure that the product is square, and in particular such a number must be a multiple of 5. Recall that multiples of 5 can only end in the digits 0, 5. Since the digit 5 has been taken, such a number must end in the digit 0. Finally since all the numbers are between 1 and 59 and the digits 4, 5 have been taken, the only permissible numbers that use the digit 0 are 10, 20 and 30. $\square$

Claim 3: The chosen number that uses the digit 7, must be 27.

Proof: Let x be the chosen number that uses the digit 7. Thanks to claim 1, we need only demonstrate that $x \neq 7$. We proceed by way of contradiction. Suppose that $x = 7$. Another one of the chosen numbers must have an odd 7-adic valuation in order to ensure that the product is square, and in particular such a number must be a multiple of 7. The only other multiples of 7 in the given range that don't use the digits 4, 5, 7 are 21, 28.

Suppose that the other multiple of 7 is 21. Since the digits 1, 2 are taken another one of the chosen numbers must be 30 as a consequence of claim 2. The remaining digits are 6, 8, 9, and we need to choose two more numbers. It is easy to check that one of the numbers must have two digits and the other one must have one digit. However, the two digit number is at least $68 > 59$, a contradiction.

Suppose that the other multiple of 7 is 28. Again due to claim 2, one of the other numbers must be from the set $\{10, 30\}$. If the number 10 is chosen then the remaining digits are 3, 6, 9. There are two numbers left to choose. Since $v_2(45 \cdot 7 \cdot 28 \cdot 10) = 3$, exactly one of the last two numbers must have an odd 2-adic valuation. The only number that has an odd 2-adic valuation is 6, so the number 6 has to be chosen also. Thus the complete set of chosen numbers is $\{45, 7, 28, 10, 6, 39\}$. However since $39 = 3 \cdot 13$ and this is the only multiple of 13 amongst the six numbers, the product isn't a perfect square, a contradiction. $\square$

So we must have chosen the numbers 45, 27. Due to claim 2, we must've also chosen one of the numbers 10, 30. If we choose the number 10 then the remaining digits are 3, 6, 8, 9, and we have to choose three more numbers. It is easy to check that one of the numbers must have two digits, and the other two numbers must have one digit each. The two digit number must come from the set $\{36, 38, 39\}$. Now one can check that if we choose the number $38 = 2 \cdot 19$ then the product won't be a perfect square since there is no way to get an additional factor of 19, similarly if we choose the number $39 = 3 \cdot 13$ then the product won't be a perfect square since there is no way to get an additional factor of 13. Thus the two digit number is 36, and the complete set of chosen numbers is $\{45, 27, 10, 36, 8, 9\}$. But since $v_3(45 \cdot 27 \cdot 10 \cdot 36 \cdot 8 \cdot 9) = 9$, the product isn't a perfect square. Thus we cannot choose the number 10.

Thus we must have chosen the numbers 45, 27, 30. The remaining digits are 1, 6, 8, 9 and we need to choose three more numbers. Again we must have 1 two digit number and 2 one digit numbers. The two digit number is one of 16, 18, 19, and from here it is easy to check case by case that the complete set of chosen number is $\{45, 27, 30, 16, 8, 9\}$, which has product $45 \cdot 27 \cdot 30 \cdot 16 \cdot 8 \cdot 9 = (6480)^2$. ■

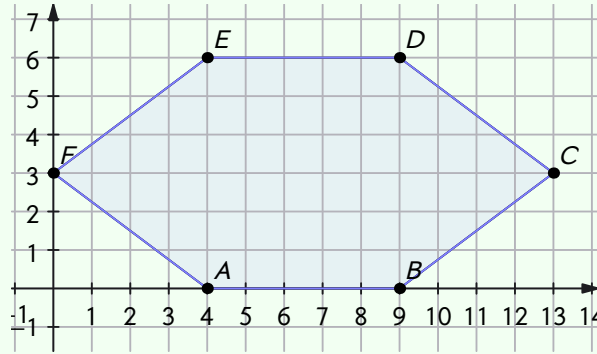
Problem 6

An *equilateral* shape is one where all the sides have the same length.

- (a) Construct an equilateral hexagon of side length 5 such that all the vertices have integer coordinates.
- (b) Is it possible to construct an equilateral pentagon of side length 5 where all the vertices have integer coordinates?

Solution

(a) Consider the following construction.



Each pair of adjacent points differs by the vector $\langle \pm 5, 0 \rangle$ or by the vector $\langle \pm 4, \pm 3 \rangle$. Since both vectors have magnitude 5 this construction is valid. $\square$

(b) Call an equilateral shape *primitive* if all its vertices lie on lattice points. In what follows if we have $\pm$ in both the x and y component of a point/vector we assume that each sign is chosen independently.

Claim 1: On a primitive shape with side length 5, adjacent vertices must differ by one of the vectors $\langle \pm 5, 0 \rangle$ or $\langle \pm 3, \pm 4 \rangle$ and their permutations.

Proof: Let P and Q be adjacent vertices of a primitive shape. Without loss of generality we can let one of the vertices lie on the origin, because we can translate the primitive shape in such a way that this occurs. Let $P = (0, 0)$, $Q = (a, b)$ for some $a, b \in \mathbb{Z}$. Because the primitive shape has side length 5, we have

$$a^2 + b^2 = 5^2.$$

Now one can test the squares $a^2 = 0, 1, 4$ and we find that (a, b) is a permutation of $(\pm 5, 0)$ or $(\pm 3, \pm 4)$, from which the desired result follows. $\square$

Say that a vector is of type I if it is of the form $\langle \pm 5, 0 \rangle$ or $\langle \pm 3, \pm 4 \rangle$, say that a vector is of type II if it is of the form $\langle \pm 4, \pm 3 \rangle$ or $\langle 0, \pm 5 \rangle$. Notice that type I vectors have an odd x component and an even y component and vice versa for type II vectors.

Suppose that a primitive pentagon of side length 5 does indeed exist. Let it have vertices $P_1, P_2, \dots, P_5$. We must have

$$\vec{V} := \overrightarrow{P_1 P_2} + \overrightarrow{P_2 P_3} + \dots + \overrightarrow{P_5 P_1} = \vec{0}$$

Suppose that among $\overrightarrow{P_1 P_2}, \overrightarrow{P_2 P_3}, \dots, \overrightarrow{P_5 P_1}$, there are n vectors of type I, and therefore $(5 - n)$ vectors of type II. Because $5 - n \equiv 1 + n \pmod{2}$, n and $(5 - n)$ have different parities. If there is an odd number

of type I vectors then $\vec{V}$ has an odd x component, and if there is an odd number of type II vectors then $\vec{V}$ has an odd y component. In both cases it is the case that $\vec{V}$ cannot be the zero vector, contradiction. ■

Comments: Frankly a very nice problem, finding the invariant that one of the components of the vector $\overrightarrow{P_1P_2} + \overrightarrow{P_2P_3} + \cdots + \overrightarrow{P_5P_1}$ must be odd allows us to finish the contradiction nicely.

Problem 7

A *quirky* sequence is a sequence of positive integers $a_1, a_2, \dots$ satisfying the following properties:

- $a_1 = 1$;
- $a_{i+1} > a_i + 1$ for every $i \geq 1$, and
- $a_i \pm a_j$ is composite for every i, j with $i > j$.

(a) Construct a quirky sequence.

(b) Given a set of numbers $a_1, a_2, \dots, a_n$ satisfying the quirky properties, can this always be extended into an infinite quirky sequence?

Solution

(a) Define $a_k = 5^{k-1}$. The first two conditions are clearly satisfied. Since the two smallest numbers in the sequence are $a_1 = 1, a_2 = 5$ we have

$$a_i + a_j \geq 1 + 5 = 6$$

$$a_i - a_j \leq 1 - 5 = -4$$

Notice that a_k consists of only odd terms. Now since $a_i \pm a_j$ is the sum of two odd integers, it is even. This coupled with the fact that $|a_i \pm a_j| \geq 4$ implies that the quantity $a_i \pm a_j$ is in fact composite. $\square$

(b) Suppose $a_1, a_2, \dots, a_n$ is a quirky sequence. Let $c_1, c_2, \dots, c_{2n}$ be composite positive integers such that they are pairwise coprime. If we can find a solution $a_{n+1} > a_n + 1$ to the system

$$a_{n+1} + a_1 \equiv 0 \pmod{c_1}$$

$$\vdots$$

$$a_{n+1} + a_n \equiv 0 \pmod{c_n}$$

$$a_{n+1} - a_1 \equiv 0 \pmod{c_{n+1}}$$

$$\vdots$$

$$a_{n+1} - a_n \equiv 0 \pmod{c_{2n}}$$

then the sequence $a_1, a_2, \dots, a_{n+1}$ is also quirky. However such a solution is guaranteed due to the Chinese remainder theorem. Now by iteratively repeating this argument we can keep on introducing new terms into the sequence, and so indeed we can create an infinite quirky sequence. $\blacksquare$

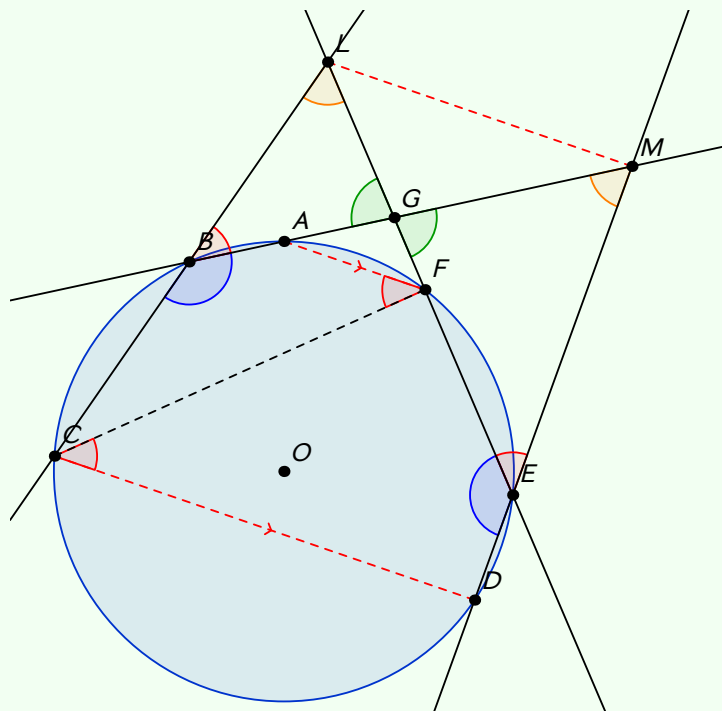
Problem 8

$ABCDEF$ is a cyclic hexagon in which AF and CD are parallel. The lines BC and EF intersect at L , and the lines AB and DE intersect at M .

Prove that LM is parallel to CD .

Solution

The argument splits into two parts. The following diagram gives an overview of the first part of our argument.



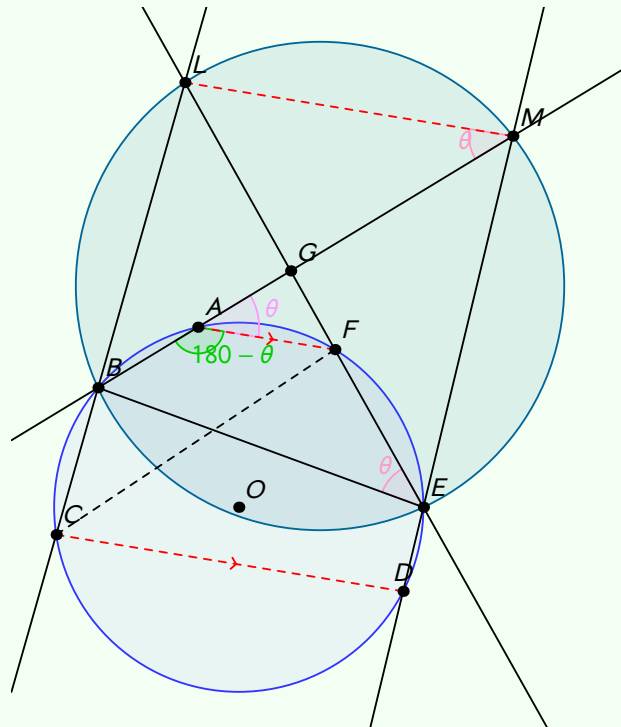
Claim 1: $\triangle BGL \sim \triangle EGM$.

Proof: Owing to the alternate angle theorem, $\angle AFC = \angle FCD := \theta$. Now since opposite angles in a cyclic quadrilateral sum to 180° , we have $\angle FBC = \angle FED = 180 - \theta$. But now since angles on a straight line sum to 180° we deduce $\angle LBG = \angle GEM = \theta$. We also have $\angle BGL = \angle EGM$ due to vertically opposite angles being equal. Now since $\triangle BGL$ and $\triangle EGM$ have two equal angles, their third angles' must be equal also and so $\triangle BGL \sim \triangle EGM$ by AAA criterion. $\square$

Claim 2: Quadrilateral $LMEB$ is cyclic.

Proof: We note that $\angle LBM = \angle LEM$, and so the line segment LM subtends two equal angles at points B and E . By the converse of the "Angles subtending the same segment are equal" theorem, we can deduce that $LMEB$ are concyclic. $\square$

The following diagram gives an overview of the second part of our argument.



Let $\angle GAF = \theta$. Since B, A and G lie on a straight line, $\angle BAF = 180 - \theta$.

Now consider the cyclic quadrilateral $ABEF$. Since opposite angles in a cyclic quadrilateral sum to 180° we have $\angle BEL = \theta$.

Now look at the circle through $LMEB$. The segment LB subtends the points M and E , so by the "Angles subtending the same segment are equal" theorem, $\angle BEL = \angle LMB = \theta$.

Now finally by the converse of the "Alternate angles are equal" theorem it must be the case that $AF \parallel LM$. ■